

## QATCH openQCM Real-Time SW Change Control Document

| Release Version | Date       | Author  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>v2.7b1</b>   | 05/06/2026 | A. Ross | <p><b>SWEEP PERFORMANCE:</b><br/> <b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b></p> <ul style="list-style-type: none"> <li>- Initial beta version for the v2.7x branch</li> <li>- Updated build information and updater client</li> <li>- New run status ui</li> <li>- Drying mode indicator</li> <li>- Multiprocessing improvements</li> <li>- Always display 10 points in analyze</li> <li>- Add CSV export with selectable columns</li> <li>- Multiple unhandled runtime exceptions during analyze</li> <li>- Version 2.0 VisQ.AI UI and Backend</li> <li>- Restore fill line filtering and tighten point threshold</li> <li>- Visqai performance optimizations</li> <li>- Adjust heating PID parameters for performance</li> <li>- Add new YOLO model assets</li> <li>- Erroneous difference curve inversion triggering</li> <li>- Head tail spike clipping, fix thread degradation</li> <li>- Modify fill classifier status messages</li> <li>- Simplify/standardize dry detection messaging</li> <li>- Warning indicators for long fills</li> <li>- Move dry detection tolerances to constants.py</li> <li>- Add db_sync module for app.db</li> <li>- Refine dry detection and extend fill warning workflows</li> <li>- Add fill complete message to primary plot</li> <li>- Add 3min initial fill timeout</li> <li>- Add mutable buffer pH to Run Info for VisQ.AI</li> <li>- Manifest driven multi-module loading for .visq files</li> <li>- Automatic cartridge lock with manual override</li> <li>- Resolving CBD encryption bug and downstream issues</li> <li>- Missing average viscosity and stdev in output files</li> <li>- Deprecate embedded progress bars in run and analyze</li> <li>- Suppress redundant session expiration prompts on idle</li> <li>- Missing _stopFreq attribute error when peak way off</li> <li>- Fix peak out of frequency range in serial</li> <li>- Fix possible IndexError in Analyze.run()</li> <li>- Refactor _update_plot() and upgrade performance</li> <li>- Various Analyze Module Improvements</li> <li>- Bug fixes for DryingDetection and FillClassifier stalls</li> <li>- Add Recover tab to Export UI with placeholder msg</li> <li>- Fix drop effect under-correction and reduce noise jitter</li> <li>- Concentration-based surface tension correct formula</li> <li>- Introduce new fill method for visco and shear calculation</li> <li>- Embed progress bars for initialize, qmodel, and analyze</li> <li>- Optimize load run sequence in Analyze</li> <li>- Restrict analysis points to 60% only by removing 80%</li> <li>- Resolve y-axis label and text overlap issues in analysis</li> </ul> |

# SW Changes

|               |            |         |                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>v2.7r1</b> | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Same change log as v2.7b1</li> </ul>                                                                                                                                                                                                                       |
| <b>v2.7b2</b> | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Initial beta version for the v2.7x branch</li> <li>- Updated build information</li> <li>- Recover DB if corrupted on application start-up</li> <li>- Fix updater so r-builds can switch from v2.6x to v2.7x</li> </ul>                                                                          |
| <b>v2.7r2</b> | 05/06/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Same change log as v2.7b2</li> </ul>                                                                                                                                                                                                                       |
| <b>v2.7b3</b> | 05/19/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Proposed fix for Live Fill Detection thread issue</li> <li>- Show "Stopped" when user clicks "Stop" button</li> <li>- Fix label overflow where long text lines are clipped</li> </ul>                                                                      |
| <b>v2.7r3</b> | 05/20/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- Proposed fix for Live Fill Detection thread issue</li> <li>- Show "Stopped" when user clicks "Stop" button</li> <li>- Fix label overflow where long text lines are clipped</li> </ul>                                                                      |
| <b>v2.7b4</b> | 05/22/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><ul style="list-style-type: none"> <li>- Updated build information</li> <li>- rate limiting to 0.05s across the board</li> <li>- fix rolling behavior for fast and long runs</li> <li>- labels in status overlay unreadable on smaller screens</li> <li>- Refactor 'RingBuffer' and Patching FillClassifier/DryDetection Bugs</li> </ul> |

## SW Changes

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| <b>v2.7r4</b> | 05/22/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><br>- Updated build information<br>- rate limiting to 0.05s across the board<br>- fix rolling behavior for fast and long runs<br>- labels in status overlay unreadable on smaller screens<br>- Refactor `RingBuffer` and Patching<br>FillClassifier/DryDetection Bugs                                                                                                 |
| <b>v2.7b5</b> | 07/09/2026 | A. Ross | <b>SWEEP PERFORMANCE:</b><br><b>FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: &lt;2s</b><br><br>- Updated build information<br>- Drop effect improvements<br>- Adding QModel Onyx<br>- Fix #419: Initial Fill Trendline Check Uses Single Point<br>- Prediction-thread leaks and crashes in VisQ.AI mode<br>- Fix memory leaks in Analyze.py to boost performance<br>- Fix Temperature Control Buffer Bug<br>- Adding tools UI with Integrated Online Calculators |
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